

INFRAGARD HOUSTON AI-CSC · MAY 14, 2026

AI Agent Teams for **Critical Infrastructure**

Practical workflows, security guardrails, and governed automation

80 min

BRAD BLOCK + Q&A

Virtual

INFRAGARD HOUSTON

Practical

WORKFLOW PATTERNS

 11:30 AM - 1:00 PM Central

ABOUT BRAD GROUX

CEO, Digital Meld

Enterprise technologist, OpenClaw contributor, and builder of agent-driven workflows for real operating teams.

 linkedin.com/in/bradgroux

 @bradgroux on Twitter

 github.com/bradgroux

 youtube.com/bradgroux

 linktr.ee/digitalmeld



25+ years in enterprise IT

NASA contractor background, Microsoft platforms, large-scale deployments, practical operations.



OpenClaw contributor and maintainer

Open-source agent workflows, Microsoft Teams integration, and operator-oriented automation.



Creator, Veritas Kanban

Task telemetry, agent orchestration, and delivery management for real build loops.



Start Small, Think Big

Community, training, office hours, and practical AI adoption for builders and operators.

FROM AWARENESS TO PRACTICE

The last talk answered what agents are. Today is about using them safely.

Prior session

Agent basics, enterprise platforms, guardrails, and critical infrastructure threat surfaces.

This session

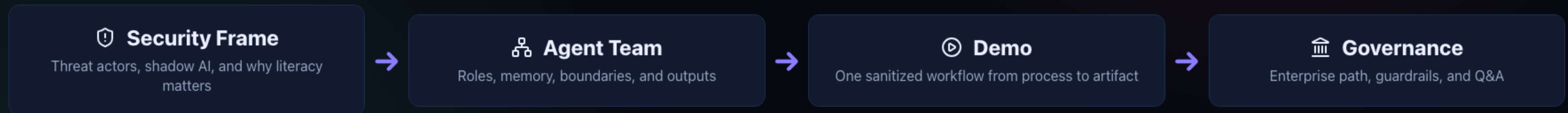
Practical agent team design, Markdown memory, verification, and a repeatable workflow demo.

Outcome

One low-risk process, one memory file, one agent team, one guardrail checklist, one next step.

80-MINUTE WORKING SESSION

Macro, micro, macro.



WHAT WE ARE CARRYING FORWARD

Agents are useful because they combine context, tools, memory, and review.

LLM brain

Reasoning engine that needs clear context and boundaries.

Tools

Files, APIs, shell commands, search, calendars, tickets, docs, and other systems.

Memory

Markdown, task history, vector stores, repo files, and reusable project knowledge.

The risk is not theoretical

- ✓ Shadow AI is already inside organizations.
- ✓ Prompt injection changes how data can attack tools.
- ✓ Agent credentials become new privileged identities.
- ✓ Logs and review decide whether you can govern it.

THE SHIFT

A chatbot answers. An agent team works a process.

 Chatbot

Single-turn responses. Weak continuity. Good for quick drafting and explanation.

 Agent

Goal-directed work. Tools, files, memory, steps, and reporting back.

 Agent Team

Specialized roles working the same process from different angles.

If you can write an SOP for it, an agent can usually help map it, check it, draft it, or improve it.

FIVE USEFUL ROLES

Do not ask one generic assistant to be the whole company.

 Process Mapper

Turns messy work into steps, owners, inputs, outputs, decisions, and handoffs.

 Security Reviewer

Finds sensitive data, access risks, compliance issues, approval gates, and audit needs.

 Documentation Writer

Drafts SOPs, checklists, runbooks, templates, and explainers.

 Implementation Planner

Breaks improvements into tasks, gates, rollback criteria, and next actions.

 Automation Scout

Suggests safe automation opportunities ranked by impact, effort, and risk.

 Human Orchestrator

Sets scope, approves sensitive work, verifies outputs, and owns the decision.

WHY MARKDOWN MATTERS

Plain text is the shared memory layer.

Markdown is readable by people, parseable by agents, renderable by GitHub, and easy to version.

- 👁 Humans can inspect it.
- 🔑 Teams can version it.
- 📄 Agents can reuse it.
- 🛡 Reviewers can audit it.

```
# Workflow Memory
```

```
## Goal
```

```
Improve a low-risk process.
```

```
## Data Boundary
```

```
Use sanitized examples only.
```

```
## Agent Team
```

- Process Mapper
- Security Reviewer
- Documentation Writer

```
## Verification
```

```
Human reviews before use.
```


OPERATING MODEL

The workflow is stable even when the tool changes.



SECURITY BEFORE AUTOMATION

Use sanitized examples. Treat output as draft work.

⊘ Do not paste

- ✗ Passwords, keys, tokens, private keys
- ✗ Customer, patient, employee, legal, or regulated records
- ✗ Internal IPs, network diagrams, firewall rules
- ✗ Unpatched vulnerabilities or production control details

☑ Do instead

- ✓ Use mock names and sanitized facts
- ✓ Keep first workflows read-only or draft-only
- ✓ Review every output before use
- ✓ Keep humans in approval loops

PRACTICAL DEMO

Field report to customer-ready summary.

 Current state

Field notes arrive in different formats. Missing data is discovered late. Customer updates vary by coordinator.

 First useful output

A field-report checklist, customer-summary template, guardrail list, and safe pilot plan.

Why this scenario works

It translates across energy, water, healthcare operations, logistics, security, and infrastructure without needing private data.

STEP 1

Create memory before asking for output.

The agent should start from stable context, not scattered chat fragments.

```
# InfraGard Demo Memory
```

```
## Workflow
```

```
Field technicians collect site observations.
```

```
A coordinator creates customer-safe summaries.
```

```
## Current Pain
```

```
- inconsistent notes
```

```
- missing data found late
```

```
- slow reporting
```

```
## Data Boundary
```

```
Use mock site names and mock measurements.
```

```
No customer names, photos, locations, or regulated data.
```


STEP 2

Good agents ask questions before they plan.

Read this Markdown memory.

Before proposing a solution, ask me three clarifying questions that would help you map this workflow safely.

Do not ask for secrets, customer data, regulated data, internal network details, or private operational details.

🔍 Scope

What exact output should this workflow produce?

👤 Owners

Who captures, reviews, approves, and sends the report?

🛡️ Boundary

Which data must stay out of the tool?

STEP 3

Map the workflow before automating anything.

The first win is clarity: steps, owners, inputs, outputs, handoffs, and missing information.

 Steps

Capture notes, review completeness, draft summary, approve, send.

 Handoffs

Technician to coordinator to reviewer to customer contact.

 Gaps

Missing measurements, unclear photos, undefined approval path.

STEP 4

Guardrails come before automation.

 Sensitive data

Customer names, exact site locations, photos, regulated records, and proprietary details stay out.

 Access

The first version is draft-only. No direct system writes, no customer sends, no production control.

 Approval

Human reviewer approves facts, safety language, and customer-ready wording.

STEP 5

Turn the process into something people can use.

A practical artifact beats an impressive abstract answer.

 Field report checklist

- ✓ Site alias and date
- ✓ Required measurements captured
- ✓ Exceptions or missing data flagged
- ✓ Reviewer approval recorded
- ✓ Customer summary generated from approved facts

STEP 6

The safest first experiment is usually draft-only.

▶ First pilot

Use five sanitized historical examples. Generate summaries. Compare against human-written reports. Keep the agent out of production systems.

⚠ Stop conditions

Stop if the agent invents measurements, loses safety caveats, asks for sensitive data, or suggests bypassing review.

Measure success

Less missing data, faster draft turnaround, consistent customer wording, and fewer review corrections.

STEP 7

Verification is part of the workflow.

Do not bolt it on later. Ask the agent to challenge its own output, then use human judgment.

Act as a skeptical reviewer.

Identify:

1. factual assumptions,
2. missing edge cases,
3. unsafe recommendations,
4. human approval points,
5. tests or checks,
6. questions before production use.

WHAT GOOD LOOKS LIKE

Useful agent output is specific, bounded, and reviewable.

🎯 Specific

Names the process, expected artifact, owners, inputs, outputs, and decision points.

🛡️ Bounded

States what data stays out, what tools are allowed, and what requires approval.

🔍 Reviewable

Lists assumptions, verification steps, stop conditions, and unanswered questions.

REAL-WORLD PATTERN

Heavy haul logistics: status clarity becomes customer value.

Manual dispatch coordination can become structured status capture, customer-safe updates, and exception escalation.

**Manual baseline**

Calls, spreadsheets, email, and inconsistent customer visibility.

**Agent-assisted workflow**

Map dispatch handoffs, standardize status fields, draft updates, flag exceptions.

**Operator lesson**

Small teams move faster when institutional knowledge becomes repeatable process.

 Field and lab data

High-value information often lives in photos, spreadsheets, emails, and informal notes.

 Agent-assisted reporting

Normalize notes, detect missing data, draft summaries, prepare customer-review packets.

 Human approval

People verify facts, safety wording, and customer-facing language before release.

REAL-WORLD PATTERN

Field data is where the boring work becomes valuable.

The leverage is not magic. It is turning messy operational data into structured, reviewed, reusable workflow.

REAL-WORLD PATTERN

Security teams should start with draft, read-only, evidence-heavy workflows.

 Low-severity triage

Summarize, classify, gather context, and draft analyst notes for review.

 Evidence packets

Collect control evidence lists, owner questions, and review checklists.

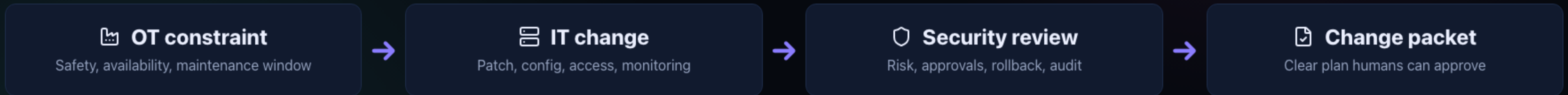
 Incident timelines

Draft timelines from sanitized notes and logs, then verify against source systems.

Agents can draft and organize. Humans own the security conclusion.

REAL-WORLD PATTERN

Agents can improve coordination without touching control systems.



ENTERPRISE GUARDRAILS

Four layers matter before broad rollout.

**Policy**

What the agent can and cannot do.

**Observability**

Logs, task IDs, actions, token usage, approvals, and review history.

**Access**

Least privilege, scoped credentials, revocation, identity separation.

**Human intervention**

Approval gates, stop conditions, and a fast circuit breaker.

SECURITY LENS

Agents introduce new ways for data and tools to be abused.

 Prompt injection

Malicious content inside data tells the agent to ignore instructions or leak information.

 Data exposure

Overbroad read access can turn ordinary agent output into exfiltration.

 Tool risk

Plugins, scripts, and integrations become supply-chain and privilege boundaries.

 Agent identity

Shared human credentials make forensics and accountability weak.

 Agent chaining

Agent-to-agent handoffs become new authorization decisions.

 Runaway scope

"While I am at it" is scope creep, cost risk, and security risk.

GOVERNED ROLLOUT

Start small, add privilege slowly.

 1. Personal literacy

Sanitized examples, local notes, no private data.

 2. Draft-only internal workflow

SOPs, summaries, checklists, plans, evidence lists.

 3. Read-only assisted workflow

Approved data sources, logging, reviewer sign-off.

 4. Governed enterprise workflow

Identity, least privilege, audit, approvals, incident response, revocation.

BAD FIRST WORKFLOWS

Some work should wait until governance exists.

External sends

Customer emails, legal notices, public posts, or regulatory statements without approval.

Production changes

Infrastructure changes, control actions, firewall updates, IAM edits, deployment changes.

Sensitive records

Patient, employee, legal, investigative, customer, or regulated data in unapproved tools.

EXAMPLE ENVIRONMENT

OpenClaw makes the workflow visible.

The point is not that every attendee must deploy OpenClaw today. The point is to see how agents use memory, tools, approvals, and logs.

**Dashboard first**

Start with visible responses before channel setup.

**Files as context**

Markdown memory, prompt packs, and project docs become reusable agent context.

**Review loop**

Inspect changes, verify outputs, and iterate from evidence.

THE TOOL WILL CHANGE

The operating model transfers across ecosystems.

<> Coding assistants

Codex, Claude Code, GitHub Copilot, and similar tools can help create files, plans, and checks.

🏢 Enterprise platforms

Microsoft, Google, NVIDIA, and others are moving toward identity, governance, and orchestration.

📁 Local/open paths

OpenClaw and open models can teach the workflow pattern without tying the lesson to one vendor.

Process, memory, role, boundary, artifact, verification. That is the durable skill.

AFTER TODAY

One low-risk workflow in 30 days.

Week 1

Write the process in Markdown. Sanitize the example. Define data boundaries.

Week 2

Run Process Mapper and Security Reviewer. Capture clarifying questions.

Week 3

Draft one artifact: SOP, checklist, summary template, or implementation plan.

Week 4

Verify with a human reviewer. Decide keep, change, or stop.

Metric

Time saved, fewer missing fields, better consistency, fewer review corrections.

Stop condition

The agent asks for sensitive data, invents facts, bypasses approval, or expands scope.

START SMALL, THINK BIG

SSTB.ai is the follow up path.

Free office hours on the SSTB Discord server, practical AI/operator playbooks, and deeper guided training.



 sstb.ai

 SSTB Discord office hours

INFRAGARD MEMBER OFFER

75% off

Use this discount code for InfraGard members:

SSTB-IGCLAW

Use it after the session for deeper training and guided implementation support.

CLICKABLE RESOURCES

Everything here is meant to be used after the session.

 **Workshop repo**
github.com/BradGroux/openclaw-dev-days

 **NIST AI RMF**
nist.gov/itl/ai-risk-management-framework

 **Prompt pack**
[event prompt-pack.md](#)

 **CISA AI resources**
cisa.gov/ai

 **Scenario cards**
[event scenario-cards.md](#)

 **MITRE ATLAS**
atlas.mitre.org

QUESTIONS

No softballs required.

Hard questions about security implications, real failures, governance, first workflows, and what I would not automate are welcome.

**Risk**

What can go wrong?

**Practice**

What should I try first?

**Governance**

How do we roll this out safely?

STAY CONNECTED

Brad Groux

Digital Meld · InfraGard Houston AI-CSC · OpenClaw contributor

-  linktr.ee/digitalmeld
-  linkedin.com/in/bradgroux
-  [@bradgroux on Twitter](https://twitter.com/bradgroux)
-  github.com/bradgroux
-  youtube.com/bradgroux



Scan for resources, slides, and contact links.